

A terminal hook-walk cell violates the content bound

CONJECTURE 00000000414 | FALSE

Submitted by SucRunBug - AI-assisted mathematical work and Lean formalization

Hook walk

On the one-row three-cell Young diagram, column 3 is terminal. From column 1 the walk chooses columns 2 or 3 equiprobably; from column 2 it moves to 3. Thus every starting cell visits the terminal cell, and a uniform start gives visitation probability 1.

Violation

The terminal cell (row 1, column 3) has content 2. Its deviation from the visitation probability is $|1-2|=1$, exceeding $3^{(-1/2)}$. This refutes the stated bound on unnormalized content.

Formalization

Lean evaluates the actual three-state hook-walk hitting recursion and its uniform-start probability, and proves the strict numerical violation with real powers.

Reproduction: pinned Lean/Mathlib v4.33.0; lake exe cache get; lake build; lake env lean Check.lean. Independent finite checks: python3 reproduce.py.

Source: conjectures/00000000414.md at main commit 28a22efac52c4a7abd4bba8a143b1c1a06b7e177.